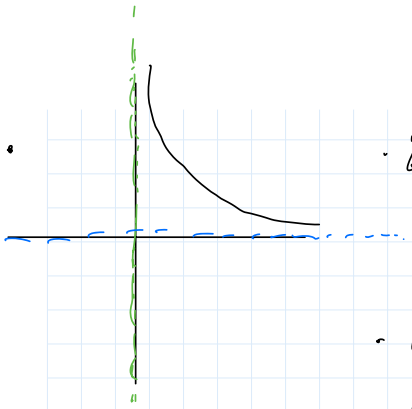
Holes and Vertical Asymptotes

$x \to 7^-$ ..... $x \to 7^+$
$a = 7$

Section 3.4B

A **vertical asymptote** is a line $x = a$ where the function grows without bound from at least one side. First simplify, but keep the original domain restrictions. Find the domain, hole, and vertical asymptote of

$$f(x) = \frac{2(x-1)}{(x-1)(x+2)}.$$



- is we know $x \to a^-$, $x \to a^+$, then what is the behavior of y? V.A.
- is we have $x \to \infty$, $x \to -\infty$ then what is the behavior of y? H.A.

- $f(x) = \frac{2}{x+2}$ is Simplified function., Domain is $x \neq 1$, $x \neq -2$.
- is a factor cancles, then it creates a hole. so $x = 1$ is whole. $x - 1 = 0$.
- is the factor stays in denominator after simplify, then it gives a vertical as.... $x = -2$ is V.A.

Approaching from the Left and Right

Use denominator signs to find the limits from each side of the vertical asymptote.

(a) $f(x) = \dfrac{1}{x-2}$ (b) $g(x) = -\dfrac{3}{(x+1)^2}$.

Explain why the two sides have opposite directions in (a), but the same direction in (b).

(a). $f(x) = 1 = 0 \cdot x^{00} + 1$

$\Rightarrow x-2=0$

$x=2$ $\Rightarrow$ $x=2$ is V.A.

- when x is approaching 2 from left hand side.

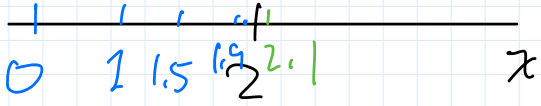
take $x=1.9$, $f(1.9) = \frac{1}{1.9-2} = \frac{1}{-0.1} = -10$

take $x=1.99$, $f(1.99) = \frac{1}{1.99-2} = \frac{1}{-0.01} = -100$

take $x=1.999$, $f(1.999) = -1000$

$\lim_{x \to 2^-} f(x) = -\infty$

$\frac{1}{\text{"number, and this number is getting smaller"}} = -\infty$



- when x is approaching 2 from r.h.s.

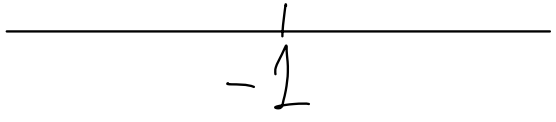
take $x=2.1$

$f(2.1) = \frac{1}{2.1-2} = \frac{1}{0.1} = 10$

$f(2.01) = \frac{1}{0.01} = 100$

$\lim_{x \to 2^+} f(x) = \infty$

(b) $g(x) = \dfrac{-3}{(x+1)^2}$, so V.A. is $x = -1$.



take $x = -0.9$

$$g(-0.9) = \frac{-3}{(-0.9+1)^2} = \frac{-3}{(0.1)^2}$$

$$g(-0.99) = \frac{-3}{(0.01)^2}$$

$\vdots$ $\quad y = -\infty$

so $\lim_{x \to -1^{\pm}} f(x) = -\infty$

Horizontal Asymptotes

A **horizontal asymptote** $y = L$ describes $f(x) \to L$ as $x \to \infty$ or $x \to -\infty$. Compare numerator and denominator degrees: smaller gives $y = 0$; equal gives the leading-coefficient ratio; larger gives no horizontal asymptote.

(a) $\dfrac{3x-1}{x^2+4}$ (b) $\dfrac{6x^2+1}{2x^2+5}$ (c) $\dfrac{x^2+1}{x-2}$.

Find each horizontal asymptote, if any. State the limits at both ends in (a) and (b).

- degree of polynomial is the order of leading term, x^2+4, degree 2, order 2.
- $\dfrac{\text{numerator}}{\text{denominator}} = \dfrac{N}{D}$

is degree $N <$ degree D, $\Rightarrow \lim_{x\to\pm\infty} y = 0$

is deg N = deg D $\Rightarrow \lim_{x\to\pm\infty} y = \dfrac{\text{lead coefi... of } N}{\text{lead coefi... of } D}$

is deg $N >$ deg D $\Rightarrow$ No H.A.

(a). $N = 3x-1$, deg $N = 1$ deg $N = 1 <$ deg $D = 2$.

$D = x^2+4$, deg $D = 2$

(c). deg $N = 2$ (V)

deg $D = 1$

No H.A.

$$\lim_{x \to \pm\infty} \frac{3x-1}{x^2+4} = 0$$

(b) $\dfrac{6x^2+1}{2x^2+5}$ $\quad \deg N = 2$ $\quad \deg D = 2$

$$\lim_{x \to \pm\infty} \frac{6x^2+2}{2x^2+5} = \frac{6}{2} = 3.$$

Combine Vertical and End Behavior

Find all vertical and horizontal asymptotes of

$$h(x) = \frac{2x-1}{x+3}.$$

Then find the limits as $x \to -3^-$, $x \to -3^+$, $x \to -\infty$, and $x \to \infty$.

$\lim_{x \to \infty} R(x) = ?$

$\lim_{x \to \pm\infty} R(x) = 2$

$\deg N = 2$

$\deg D = 2$

Two Vertical Asymptotes

For

$$r(x) = \frac{x+1}{(x-2)(x+3)},$$

find all asymptotes. Use factor signs to find the limit from each side of each vertical asymptote.